EDUCATIONAL HACKATHON PROJECT

ALUM-ECO
Smart Sustainable Cosmetics



An Educational Hackathon Project for 3rd BGU

October 23, 2026

Project Overview

ALUM-ECO combines a reusable alum stone deodorant prototype with a website that compares deodorants by cost, lifespan and packaging waste.

GENERAL OBJECTIVE

Evaluate the ecological and economic viability of alum stone deodorants.

1 Compare cost & lifespan

2 Estimate packaging waste

3 Develop comparison website

4 Create physical prototype

5 Promote informed choices

The Problem—and the Opportunity

Conventional deodorants create persistent plastic waste, while their true cost over time remains difficult to compare.

TRANSPARENT CHOICE

A simple comparison tool can turn confusing product claims into responsible decisions.

- Consumers seeking affordability
- Eco-conscious users
- Environment affected by waste
- Small sustainable businesses



One idea.
Two prototypes.

PHYSICAL

Reusable alum stone with an eco-friendly holder.

DIGITAL

Website calculating cost, lifespan and packaging-waste comparisons.

Website Functionality

INPUT

Price
Lifespan
Packaging weight

PROCESS

JavaScript calculates
30-day cost + waste

OUTPUT

Interactive charts
Estimates + comparisons

$$\text{30-day cost} = \text{price} \div \text{lifespan} \times 30$$

Clear assumptions make every result teachable—and transparent.

Technology & Materials

AI
Coding help & explanations

HTML + CSS
Structure & design

JavaScript
Calculations & interactivity

Chart library
Visual comparisons

Code editor
Development workspace

Digital scale
Weight measurement

Alum + holder
Physical prototype

Future sensor
Optional enhancement

Business Model Canvas

VALUE PROPOSITION
Transparent cost, lifespan & waste comparison

CUSTOMERS
Eco-conscious, budget-aware consumers

CHANNELS
Website, social media, fairs, sellers

RELATIONSHIPS
Education, feedback & support

ACTIVITIES
Research, prototyping, maintenance, marketing

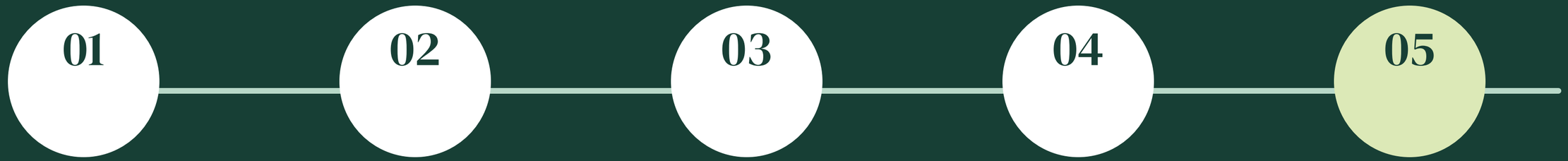
RESOURCES
Materials, website, data, team

PARTNERS
Suppliers, packagers, retailers

REVENUE
Product sales, partnerships

COSTS
Materials, packaging, development, marketing

Development Stages



Research

Surveys, competitors, pricing

Development

Website, prototype, data

Testing

Users, feedback, fixes

Launch

Social, fairs, QR codes

Evaluation

Costs, sales, impact

Marketing Strategy

INSTAGRAM
Product photography, comparisons, updates

SHORT VIDEO
Fast demos of website features

SCHOOL FAIRS
Prototype + website presentations

QR CODES
One-scan access to comparison tools

EDUCATION
Cost, lifespan and impact explained

“Compare smarter. Choose sustainably.”

Budget Overview

Alum stone	Research needed
Holder / packaging	Research needed
Digital scale	Research needed
Website development	**Free tools · minimal cost**
Marketing materials	Research needed
Other materials	Research needed

Profit = Revenue − Total costs

Expected Results

Functional website

Physical prototype

ALUM-ECO
validated concept

Cost + waste calculators

Sustainability awareness

Viability assessment

Customer feedback

Evaluation Criteria

01

Website functionality

Accurate calculations and charts

02

Cost validity

Correct formulas

03

Environmental analysis

Measured weights + clear assumptions

04

Customer satisfaction

Survey feedback

05

Economic viability

Sales and cost analysis

06

Prototype quality

Usability and presentation

Technology meets sustainability.

ALUM-ECO helps consumers make informed, eco-friendly choices. Real data and user feedback are the next steps toward validation, improvement and community impact.

Questions or collaboration?

